

WASP-39b

R-1100

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_240330 · created 2026-07-30T19:10:37+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460399.894 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.056
Transit depth (Rp/Rs)^2	3.43 +/- 2.09 [%]
Orbital Inclination (inc)	87.26 +/- 2.09
Airmass coefficient 1 (a1)	1.01 +/- 0.1
Airmass coefficient 2 (a2)	0.0 +/- 0.27
Scatter in the residuals of the lightcurve fit is	10.58 %
Best Comparison Star	#1 - [283, 350]
Optimal Aperture	2.26
Optimal Annulus	12.27
Transit Duration (day)	0.105 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.185 ± 0.056 vs 0.1457 (deviation 0.7σ)
Duration fit vs published	0.105 d vs 0.12 d (deviation 0.41σ)
Mid-time O–C	-14.5 min
Depth SNR	3.3
Residual scatter	10.58 %

Light curve

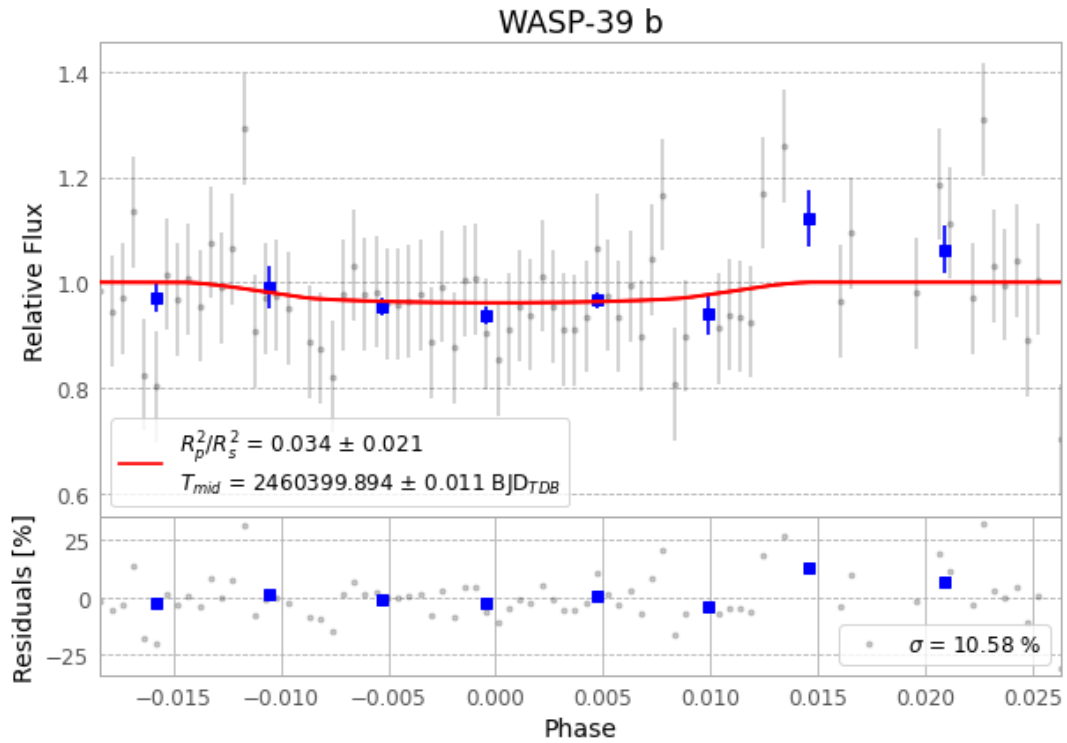


Figure 1 — FinalLightCurve_WASP-39 b_2024-03-30.png (SHA-256 e1f9a011383d1c19...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 218], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 35bd28e20735985d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

90 FITS frames (59 MB), WASP-39240330073016.FITS → WASP-39240330115715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-240330.log (f895a1fa67b6...), FinalLightCurve_WASP-39 b_2024-03-30.png (e1f9a011383d...), FinalLightCurve_WASP-39 b_2024-03-30.csv (59a2107fbc37...)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 19:10Z.